2.5 The Processor Performance Equation

- Clock cycle time $= \dfrac{1}{\text{Clock rate}}$

  ↑ sec ↑ Hz

$$\text{CPU time} = \text{Clock cycles for a program} \times \text{Clock cycle time} =$$

$$= \frac{\text{Clock cycles for a program}}{\text{Clock rate}}$$

$$\text{Clock cycles for a program} = \overbrace{\text{Instruction Count}}^{IC} \times \underbrace{\text{Clock Cycles per Instruction}}_{CPI}$$

$$\text{CPU time} = \underline{\underline{IC}} \times CPI \times \text{Clock cycle time}$$

$a * b + c$

$$\text{CPU time} = \frac{\cancel{\text{Instructions}}}{\text{Program}} \times \frac{\cancel{\text{Clock cycles}}}{\cancel{\text{Instruction}}} \times \frac{\text{Seconds}}{\cancel{\text{Clock cycle}}} = \frac{\text{Seconds}}{\text{Program}}$$

CPU time is equally dependent on all 3 parameters

=> <u>Consequence</u> :